Supplementary Material for "Anytime-Valid Identification of Pareto-Optimal Alternatives from Noisy Simulations"

References to sections, equations, theorems and figures without the prefix S are to the main article.

S1 Proof of Proposition 1 and a remark on cycles with two objectives

Throughout, $\varepsilon_i = \varepsilon$, every box contains its mean and every box has width at most η in every objective. If $\boldsymbol{\mu}_j - \boldsymbol{\mu}_k \le (\varepsilon - 2\eta)\mathbf{1}$ componentwise, then

$$\boldsymbol{U}_j - \boldsymbol{L}_k \le (\boldsymbol{\mu}_j + \eta\mathbf{1}) - (\boldsymbol{\mu}_k - \eta\mathbf{1}) \le \varepsilon\mathbf{1},$$

so j certifiably $\boldsymbol{\varepsilon}$-dominates k. Conversely, if $\mu_{j,i} - \mu_{k,i} > \varepsilon$ for some i, then $U_{j,i} - L_{k,i} \ge \mu_{j,i} - \mu_{k,i} > \varepsilon$ and j is never certified to $\boldsymbol{\varepsilon}$-dominate k. In each part below, every ordered pair falls into one of these two cases, and η is half the smallest slack of the first case.

(a) The three mean vectors are distinct permutations of (a, b, c) and have equal coordinate sums, so none dominates another and $\mathcal{P} = \{1, 2, 3\}$. For the pair $(1, 2)$, $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = (a - c,\ b - a,\ c - b)$, whose largest component is $\max(b - a, c - b) < \varepsilon$; by cyclic symmetry the same holds for $(2, 3)$ and $(3, 1)$. With $\eta = \{\varepsilon - \max(b - a, c - b)\}/2$, alternative 1 certifiably $\boldsymbol{\varepsilon}$-dominates 2, 2 dominates 3 and 3 dominates 1. In the reverse direction, $\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1 = (c - a,\ a - b,\ b - c)$ has first component $c - a > \varepsilon$, so 2 never $\boldsymbol{\varepsilon}$-dominates 1, and likewise for the other reversed pairs. There is no tie, and each alternative is certifiably $\boldsymbol{\varepsilon}$-dominated by another: $\boldsymbol{\varepsilon}$-elimination discards all three, whatever the tie-break.

(b) The points are mutually nondominated because $x_1 < x_2 < x_3$ and $y_1 > y_2 > y_3$. The differences are

$$\begin{aligned}
\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 &= (x_1 - x_2,\ y_1 - y_2), & \boldsymbol{\mu}_2 - \boldsymbol{\mu}_1 &= (x_2 - x_1,\ y_2 - y_1),\\
\boldsymbol{\mu}_2 - \boldsymbol{\mu}_3 &= (x_2 - x_3,\ y_2 - y_3), & \boldsymbol{\mu}_3 - \boldsymbol{\mu}_2 &= (x_3 - x_2,\ y_3 - y_2),\\
\boldsymbol{\mu}_3 - \boldsymbol{\mu}_1 &= (x_3 - x_1,\ y_3 - y_1), & \boldsymbol{\mu}_1 - \boldsymbol{\mu}_3 &= (x_1 - x_3,\ y_1 - y_3).
\end{aligned}$$

The first five have all components below ε (negative components trivially, and $x_2 - x_1 \le x_3 - x_1 < \varepsilon$, $x_3 - x_2 < \varepsilon$, $y_1 - y_2 < \varepsilon$, $y_2 - y_3 < \varepsilon$), while $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_3$ has $y_1 - y_3 > \varepsilon$. Hence, for small η, $\{1, 2\}$ and $\{2, 3\}$ are certified $\boldsymbol{\varepsilon}$-ties and 3 certifiably and one-directionally $\boldsymbol{\varepsilon}$-dominates 1. Without a tie-break every alternative is $\boldsymbol{\varepsilon}$-dominated by another and all are discarded. With the tie-break that keeps the earlier alternative of a tie in the order $1, 2, 3$, alternative 1 is discarded by 3, alternative 2 by 1 and alternative 3 by 2. The instance of Figure 1(b) has $x = (0.30, 0.34, 0.38)$, $y = (0.50, 0.44, 0.38)$ and $\varepsilon = 0.1$, with $\eta = 0.01$.

(c) With $\varepsilon = 0.1$: $\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1 = (0.08, -0.15)$ and $\boldsymbol{\mu}_3 - \boldsymbol{\mu}_2 = (0.08, -0.15)$, so 2 certifiably $\boldsymbol{\varepsilon}$-dominates 1 and 3 dominates 2 once $\eta < 0.01$. All other ordered pairs have a component exceeding ε: $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = (-0.08, 0.15)$, $\boldsymbol{\mu}_2 - \boldsymbol{\mu}_3 = (-0.08, 0.15)$, $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_3 = (-0.16, 0.30)$, $\boldsymbol{\mu}_3 - \boldsymbol{\mu}_1 = (0.16, -0.30)$, and alternative 4 differs from every other by at least 0.2 in the first objective and 0.25 in the second, in opposite directions. Hence 1 and 2 are discarded and 3 and 4 survive. Neither covers 1: $\mu_{3,1} = 0.46 > 0.30 + 0.1$ and $\mu_{4,2} = 0.95 > 0.70 + 0.1$. The four points are mutually nondominated, so $\mathcal{P} = \{1, 2, 3, 4\}$.

Remark: no one-directional cycle of length three with two objectives. Let three mutually nondominated alternatives satisfy $x_1 < x_2 < x_3$ and $y_1 > y_2 > y_3$, and suppose their $\boldsymbol{\varepsilon}$-dominance relations form a directed cycle in which no pair is a tie. There are two orientations. If the cycle is $1 \to 2 \to 3 \to 1$, then $3 \to 1$ requires $x_3 \le x_1 + \varepsilon$; since $x_2 \le x_3$ and $y_2 \le y_1$, alternative 2 also $\boldsymbol{\varepsilon}$-dominates 1, so $\{1, 2\}$ is a tie, a contradiction. If the cycle is $1 \to 3 \to 2 \to 1$, then $1 \to 3$ requires $y_1 \le y_3 + \varepsilon$; since $y_2 \le y_1$ and $x_2 \le x_3$, alternative 2 $\boldsymbol{\varepsilon}$-dominates 3, so $\{2, 3\}$ is a tie, again a contradiction. Part (b) of the proposition shows that the tie-break re-creates the failure.

S2 Width of the Hoeffding confidence sequence and planning horizons

The predictable plug-in Hoeffding sequence of Waudby-Smith & Ramdas (2024, Proposition 1) at level $\alpha' = \alpha/(Km)$ uses $\lambda_t = \min\{1, \sqrt{8A/(t \log(t+1))}\}$ with $A = \log(2/\alpha')$, and after n evaluations reports the interval centred at $\sum_t \lambda_t X_t / \sum_t \lambda_t$ with half-width

$$h(n) = \frac{A + \sum_{t \le n} \lambda_t^2 / 8}{\sum_{t \le n} \lambda_t}.$$

The half-width does not depend on the data. As $n \to \infty$, $\lambda_t = \sqrt{8A/(t \log(t+1))}$ for all but finitely many t, $\sum_{t \le n} \lambda_t \sim 2\sqrt{8An/\log n}$ and $\sum_{t \le n} \lambda_t^2/8 = A \sum_{t \le n} \{t \log(t+1)\}^{-1} \sim A \log\log n$. Hence

$$h(n) \sim \Big(\frac{A \log n}{32\, n}\Big)^{1/2} (1 + \log\log n),$$

the box width is at most $w(n) = 2h(n)$, and inverting gives

$$n^*(x) = O\{x^{-2} A \log(1/x) (\log\log(1/x))^2\} \qquad (x \to 0).$$

Table S1 reports the exact values of $n^*(\varepsilon/4)$ and of the horizon $K\{n^*(\varepsilon/4) + b\}$ of Theorem 2, computed from the formula above with $b = 5$ and $\alpha = 0.05$.

S3 Software

The procedure is implemented in the R package paretoinfer. A minimal identification with a user-supplied simulator `simulate(alternative, n)`, which returns n evaluations of one alternative as a data frame with one column per objective, is

Table S1: Sufficient budgets of Theorem 2 for the predictable plug-in Hoeffding sequence, $\alpha = 0.05$, $b = 5$: number of evaluations per alternative $n^*(\varepsilon/4)$ and total horizon $K\{n^*(\varepsilon/4)+b\}$.

ε	m	K	$n^*(\varepsilon/4)$	horizon
0.025	2	5	2,085,690	10,428,475
0.025	2	10	2,299,452	22,994,570
0.025	2	20	2,511,220	50,224,500
0.025	3	5	2,210,833	11,054,190
0.025	3	10	2,423,677	24,236,820
0.025	3	20	2,634,526	52,690,620
0.050	2	5	419,318	2,096,615
0.050	2	10	462,845	4,628,500
0.050	2	20	506,005	10,120,200
0.050	3	5	444,794	2,223,995
0.050	3	10	488,159	4,881,640
0.050	3	20	531,155	10,623,200
0.100	2	5	81,299	406,520
0.100	2	10	89,895	899,000
0.100	2	20	98,430	1,968,700
0.100	3	5	86,328	431,665
0.100	3	10	94,900	949,050
0.100	3	20	103,408	2,068,260

```
library(paretoinfer)
design <- pareto_design(alternatives = c("A1", "A2", "A3", "A4"),
                        objectives = c("y1", "y2"), bounds = c(0, 1),
                        alpha = 0.05, epsilon = 0.05, cover = TRUE,
                        max_evaluations = 1e5)
state <- pareto_run(design, simulate, batch = 5, initial = 5)
plausible_pareto_set(state)  # sets and representatives
pareto_report(state)          # design, sets, boxes, assumptions, versions
autoplot(state)               # confidence boxes coloured by status
```

When evaluations are produced elsewhere, `initialize_pareto()`, `update_objectives()` and `choose_next_evaluation()` drive the same loop by hand. Evaluations outside the declared bounds, missing values and unknown alternatives are refused with an error rather than repaired. A stopped identification accepts no further evaluations, so that a report always refers to a design fixed in advance.

S4 Additional results of Study 2

Table S2 reports, for every cell of Study 2, the median and the quartiles of the number of evaluations until stopping for the open-pair and the uniform allocation, and, for ε-discarding with its own allocation, the percentage of replications that stopped within 3×10^4 evaluations and the percentage with frontier loss. The first nine rows are the main grid of Figure 2. No replication of any method produced a false statement.

Table S2: Study 2: median (quartiles) of the number of evaluations; 30 instances per cell, budget 3×10^4.

K	m	ε	ρ	sequence	open-pair	uniform	ε-disc. stop / loss (%)
5	2	0.025	0.0	betting	1,568 (1,199, 3,040)	2,138 (1,820, 4,421)	63 / 0
10	2	0.025	0.0	betting	4,390 (3,701, 5,884)	6,640 (5,870, 8,741)	33 / 0
20	2	0.025	0.0	betting	12,175 (11,341, 15,310)	16,682 (15,344, 20,986)	0 / 0
5	2	0.050	0.0	betting	1,250 (1,001, 1,779)	2,240 (1,294, 3,565)	90 / 0
10	2	0.050	0.0	betting	3,612 (2,986, 4,260)	4,960 (3,658, 6,935)	53 / 3
20	2	0.050	0.0	betting	8,388 (7,571, 10,562)	13,512 (8,621, 15,061)	23 / 17
5	2	0.100	0.0	betting	890 (824, 991)	1,092 (942, 1,342)	97 / 3
10	2	0.100	0.0	betting	2,362 (2,075, 2,499)	2,885 (2,649, 3,099)	83 / 3
20	2	0.100	0.0	betting	5,020 (4,812, 5,422)	6,225 (6,092, 6,600)	73 / 43
10	3	0.050	0.0	betting	5,018 (4,102, 5,891)	7,835 (6,521, 8,630)	
10	2	0.050	-0.6	betting	4,135 (3,294, 5,141)	4,795 (3,015, 6,604)	
10	2	0.050	+0.6	betting	3,972 (3,289, 4,672)	4,962 (3,880, 6,200)	
10	2	0.050	0.0	emp. Bernstein	6,455 (4,945, 8,450)	6,572 (5,535, 10,350)	
10	2	0.050	0.0	Hoeffding	30,000 (30,000, 30,000)	30,015 (30,000, 30,029)	

S5 Details of the applications

Binomial intervals. The eight intervals have nominal level 0.95 and are truncated to $[0, 1]$. Wald, Wilson and Agresti–Coull follow Agresti & Coull (1998); Wilson-cc is method 4 of Newcombe (1998); Clopper–Pearson uses beta quantiles (Clopper & Pearson 1934); Jeffreys is the equal-tailed $\mathrm{Beta}(x + 1/2, n - x + 1/2)$ interval with the modification of Brown et al. (2001) at $x = 0$ and $x = n$; the arcsine interval uses $\tilde{p} = (x + 3/8)/(n + 3/4)$; and the likelihood-ratio interval inverts the binomial deviance at $\chi^2_{1,0.95}$. With $p \sim U(0, 1)$, X is uniform on $\{0, \ldots, n\}$ and $p \mid X = x \sim \mathrm{Beta}(x+1, n-x+1)$, so the average coverage of an interval $[l(x), u(x)]$ is $(n+1)^{-1} \sum_x \{F_x(u(x)) - F_x(l(x))\}$, where F_x is that beta distribution function, and the average width is $(n + 1)^{-1} \sum_x \{u(x) - l(x)\}$. Table S3 gives these values together with the final statements of the identification of Section 6.1.

Table S3: Binomial intervals, $n = 30$: exact average miscoverage and width over $p \sim U(0, 1)$, membership of the Pareto set, number of evaluations (each the average of 100 simulated studies) and final statements after 10^5 evaluations.

Method	miscoverage (%)	width	Pareto	evaluations	statement	dominator or role
Wald	12.509	0.2663	yes	26,115	certified optimal	representative
Wilson	4.760	0.2708	yes	13,020	certified optimal	representative
Wilson-cc	2.697	0.2998	no	3,785	ε-optimal	representative
Agresti-Coull	3.986	0.2788	yes	12,100	certified optimal	representative
Clopper-Pearson	2.661	0.2990	yes	3,845	ε-optimal	
Jeffreys	4.946	0.2687	yes	14,130	certified optimal	
Arcsine	4.485	0.2739	yes	13,030	certified optimal	
Likelihood ratio	5.293	0.2684	yes	13,975	ε-optimal	

Inventory system. Write $J_t \in \{s, \ldots, S\}$ for the inventory level at the start of period t, after the arrival of any order, with $J_1 = S$. The level at the end of the period is $I_t = J_t - D_t$, the period cost is $32\mathbb{1}\{I_t < s\} + 3(S - I_t)\mathbb{1}\{I_t < s\} + \max(I_t, 0) + 5\max(-I_t, 0)$, and $J_{t+1} = S$ if $I_t < s$ and $J_{t+1} = I_t$ otherwise. The distribution of J_t is propagated exactly through the 80 periods, which gives the expected average cost and backlog fraction over

periods 21 to 80. Since $J_t \geq 10$ and $D_t \leq 50$ for every policy, the cost is at most $32 + 3(120 - 10 + 50) + 120 + 5 \cdot 40 = 832$. Table S4 lists the exact values and the final statements of Section 6.2. The four policies with $s = 40$ run out of stock in fewer than one period in 10^6, so their backlog fractions are equal for every practical purpose; the Pareto set contains $(40, 80)$ only because its backlog fraction is smaller than that of $(40, 100)$ by 4×10^{-10}. A Monte Carlo estimate from 10^5 replications per policy records no backlog for three of these policies and would declare $(40, 100)$ Pareto-optimal; this is why we use the exact values.

Table S4: (s, S) inventory policies: exact average cost per period and backlog fraction, membership of the Pareto set, number of evaluations and final statements after 10^5 evaluations.

Policy	cost	backlog	Pareto	evaluations	statement	dominator or role
(10,30)	115.61	1.46e-01	no	7,560	discarded	(20,60)
(20,40)	119.60	7.44e-02	no	2,010	discarded	(20,60)
(30,50)	127.37	3.37e-03	no	6,665	discarded	(30,70)
(40,60)	137.33	1.23e-07	no	12,245	ε-optimal	
(10,50)	109.62	2.40e-01	yes	5,595	certified optimal	representative
(20,60)	113.07	1.97e-02	yes	4,545	certified optimal	representative
(30,70)	122.62	4.91e-04	yes	6,260	certified optimal	representative
(40,80)	132.61	1.74e-08	yes	12,120	ε-optimal	
(10,70)	117.55	2.56e-01	no	1,415	discarded	(10,50)
(20,80)	117.63	5.87e-02	no	2,875	discarded	(20,60)
(30,90)	126.39	7.68e-04	no	6,430	ε-optimal	
(40,100)	136.38	1.79e-08	no	12,075	ε-optimal	
(10,90)	123.02	1.90e-01	no	1,135	discarded	(20,60)
(20,100)	124.27	7.60e-02	no	1,115	discarded	(20,60)
(30,110)	132.44	1.64e-03	no	14,970	ε-optimal	
(40,120)	142.42	4.46e-08	no	2,985	ε-optimal	

S6 Reproducibility

The archive contains the R scripts in the order in which they are run: `00_install.R` installs the paretoinfer version used here into a private library; `01_validity.R`, `01b_validity_ties.R` and `02_efficiency.R` run the simulation studies; `03_applications.R` runs the applications; `04_planning.R` computes the horizons of Table S1; and `fig1_failure.R`, `fig_studies.R`, `fig_apps.R` and `supp_tables.R` produce every figure and table. Shared code is in `common.R`, `app_binomial_methods.R` and `app_inventory_model.R`. Every script sets its seeds and the studies use parallel streams of the L'Ecuyer generator; because the streams are assigned to worker processes, the results are reproduced exactly with the same number of cores, four in our runs. The number of replications is read from the environment variable `PI_R`, which allows a quick run with fewer replications.

References

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